$$\int\limits_0^x t^2\,dt = \frac{x^3}{3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The solutions of the equation $x = x^2 - 5x + 4$ are $x = 3 - \sqrt{5}$ and $x = \sqrt{5} + 3$. The integral is equal to

$$\int \log(x)\,dx = x\log(x) - x$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{\pi}{\tanh(\pi)}$$

$$x^3 - 4x^2 + 5x - 2 = 0 \Rightarrow x = 1, x = 2$$

$$\int_x^{x^3} \sin t\,\mathrm{d}t = \cos(x) - \cos\left(x^3\right)$$

$$\int_{-1}^{1} \log\left(x^2+1\right)\mathrm{d}x = -4 + 2\log(2) + \pi$$

$$\begin{aligned}\iint \log(xy)\,dx\,dy &= \int (x\log(xy) - x)\ dy \\ &= xy\log(xy) - 2xy\end{aligned}$$

Inline: $\arcsin\left(\frac{x}{2}\right) - \frac{\pi}{3} = 0$ Equation:

$$\sin^{-1}\left(\frac{x}{2}\right) - \frac{\pi}{3} = 0 \Rightarrow x = \sqrt{3}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The solution of the differential equation

$$2f(x) - 3\frac{d}{dx}f(x) + \frac{d^2}{dx^2}f(x) = \cos(x)$$

is

$$f(x) = C_1e^x + C_2e^{2x} - \frac{3\sin(x)}{10} + \frac{\cos(x)}{10}$$

$$\int_{-1}^{1} \tan(x)\,\mathrm{d}x = -\log(\cos(x))$$

$$\int_1^2 x \log(x) - x \, \mathrm{d}x = -\frac{9}{4} + 2\log(2)$$

$$3^3 = 27$$

$$\begin{cases} 2x - 5y = -1 \\ 3x + 4y = 10 \end{cases} \Rightarrow \quad x = 2 \quad \text{and} \quad y = 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\sum_{k=1}^{10} \frac{1}{n^2} = \frac{1968329}{1270080}$$

$$x^2 - 4x + 3 = 0 \Rightarrow x = 3$$

$$x^3 - x^2 + 3x - 10 = 0 \Rightarrow x = 2 \ , \ x = -1/2 - \sqrt{19}i/2 \ , \ x = -1/2 + \sqrt{19}i/2$$

,

$$\int_{-2}^{2} \frac{x^2}{x^2+1} \, \mathrm{d}x = x - \tan^{-1}(x)\Big|_{-2}^{2} = 4 - 2\tan^{-1}(2)$$

$$\int_{-1}^{1} x^2 \, \mathrm{d}x = \frac{x^3}{3}$$

$$\cos(2x) = 1 \Rightarrow x = 0 \quad \text{and} \quad x = \pi$$